

Polyline Editor – Problem Definition & Research Document

1. Introduction

This document presents the initial phase of the Polyline Editor project. It includes research on existing systems, identification of problems, definition of target users, system constraints, and objectives. The purpose is to establish a clear foundation for designing a user-friendly and efficient system.

2. Research on Existing Systems

AutoCAD

AutoCAD is a professional design tool used for creating complex drawings. It allows users to draw and edit polylines using commands.

Pros: Powerful features, precise control.

Cons: Complex interface, requires command knowledge, not beginner-friendly.

BrickCAD

BrickCAD is a simpler CAD tool used for basic drawing tasks.

Pros: Easier than AutoCAD, supports basic polyline editing.

Cons: Still requires multiple steps, limited usability features.

Leaflet (Polyline)

Leaflet is a web-based mapping library that supports polyline drawing on maps.

Pros: Interactive, simple visualization.

Cons: Limited editing capabilities, not focused on detailed polyline manipulation.

3. Problem Statement

Existing polyline editing tools are often complex and not user-friendly, especially for beginners. Users face difficulties in performing simple tasks like drawing, moving, and deleting points efficiently. There is a need for a simplified system that improves usability and interaction.

4. Target Audience

Beginner Users (Students): Need simple tools for quick diagram creation but struggle with complex interfaces.

Intermediate Users (Designers): Need efficient editing tools but face issues with accidental modifications and complex workflows.

5. Constraints

- Keyboard and mouse input only
- Limited canvas size
- Real-time performance required
- Support for multiple polylines
- Simplicity and ease of use

6. Objectives

- Provide a simple and intuitive interface
- Reduce user effort and complexity
- Ensure fast and efficient interaction
- Provide clear visual feedback
- Improve overall usability

7. Conclusion

This document defines the initial understanding of the problem and user needs, forming the basis for further analysis and design of the Polyline Editor system.